# Appendix S4: Summary of Findings for individual studies

## Immunologically-based tests: Crystal VC

*Table 1. Sensitivity and specificity results for individual studies reporting on Crystal VC*

| **Test: Crystal VC (Institut Pasteur/Span Diagnostics)** | | | | |
|---|---|---|---|---|
| **Intended Location: Field** | | | | |
| **Target: lipopolysaccharide antigen of V. cholerae O1 and O139** | | | | |
| **Reference test: Bacterial Culture** | | | | |
| **Sample type: Stool** | | | | |
| **Study ID** | **n** | **Enrichment details (if relevant)** | **Sensitivity** | **Specificity** |
| *Direct Samples* | | | | |
| George 2014 | 125 | | 65.6 (52.7-77.1) | 91.8 (81.9-97.3) |
| Page 2012 (lab technicians) | 256 | | 93.0 (88.3-96.6) | 85.2 (69.8-99.2) |
| Page 2012 (clinicians) | 255 | | 93.8 (89.2-97.2) [1,4] | 78.4 (59.6-98.7) [1,4] |
| Ley 2012 | 622 | | 93.1 (88.7-96.2) [4] | 49.2 (44.3-54.1) [4] |
| Mukherjee 2010 | 212 | | 91.70 | 72.90 |
| Islam 2019 | 5865 | | 72.2 (64.6-78.9) | 77.1 (75.9-78.2) |
| Sayeed 2018 | 76 | | 97.5 (87.5-99.9) [4] | 98.4 (92.0-99.9) [4] |
| Matias 2017 | 511 | | 98.6 (96.5-99.6) | 71.1 (64.6-76.9) |
| *Enriched samples* | | | | |
| George 2014 | 125 | 6h enrichment in APW | 75 (62.6-85) | 98.4 (91.2-100) |
| Bwire 2017 | 102 | 6h enrichment in APW | 98.9 (94.09-99.97) | 90 (55.5-99.75) |
| Islam 2019 | 614 | Overnight enrichment in APW | 68.3 (51.9-81.9) | 90.8 (88.1-92.9) |
| **Sample type: Water** | | | | |
| *Enriched samples* | | | | |
| Rashid 2017 [2] | 1648 | 18h enrichment in APW | 65.6 (55.2-75) | 99.6 (99.2-99.9) |
| Chakraborty 2013 | 550 | 24h incubation in APW | 87 (74.9-94.3) [1] | 100 (99-100) [1] |

| | | | | |
|---|---|---|---|---|
| **Reference test: PCR** | | | | |
| **Sample type: Stool** | | | | |
| *Direct Samples* | | | | |
| Ontweka 2016 | 100 | | 94.4 (81.3-99.3) | 79.7 (67.8-88.7) |
| Harris 2009 | 99 | | 97 (88.7-99.5) | 75 (56.2-87.9)[3] |
| *Enriched Samples* | | | | |
| Ontweka 2016 | 100 | 4-6h enrichment in APW | 86.1 (70.5-95.3) | 100 (94.4-100) |
| Debes 2016 | 673 | 24h enrichment in APW | 89.3 (71.8-97.7) [3] | 98.9 (97.8-99.6) [3] |
| **Reference test: Culture or PCR** | | | | |
| **Sample type: Stool** | | | | |
| *Direct Samples* | | | | |
| Page 2012 (lab technicians) | 256 | | 88.2 (82.6-92.4) | 88.6 (78.7-94.9) |
| Page 2012 (clinicians) | 255 | | 91.9 (87-95.4) | 82.6 (71.6-90.7) |

[1]Confidence Intervals not provided; calculated from raw numbers

[2]Combined supply and stored water

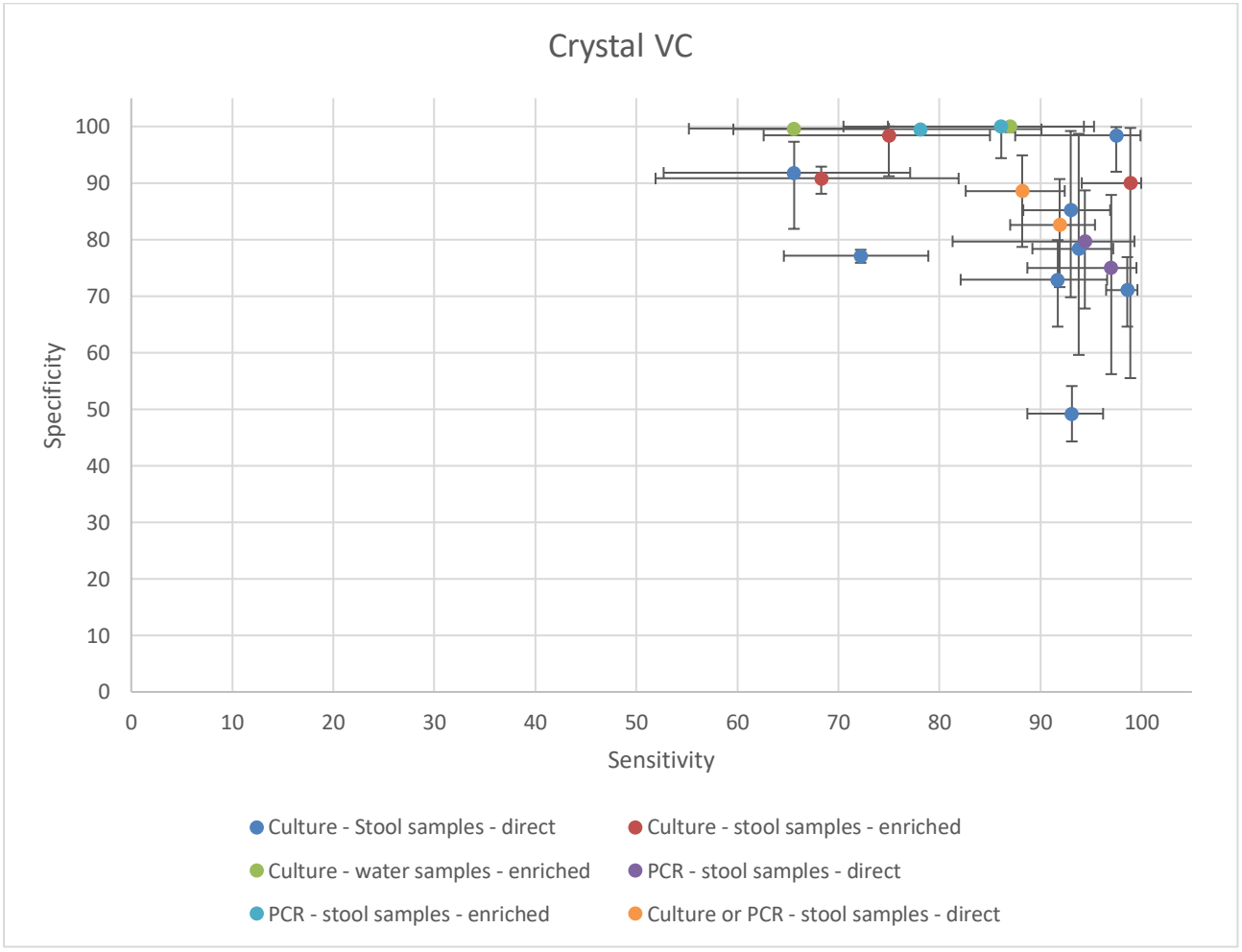
[3]Sensitivity and specificity re-calculated using raw numbers due to errors in paper

[4] Estimated by Bayesian Latent Modelling

4

5

*Figure 1. Plot of sensitivities and specificities for individual studies. Each dot represents a single study result, sub-grouped by colour according to 'reference test – sample type – enrichment status' as per the legend below. Error bars show 95% Confidence Intervals.*



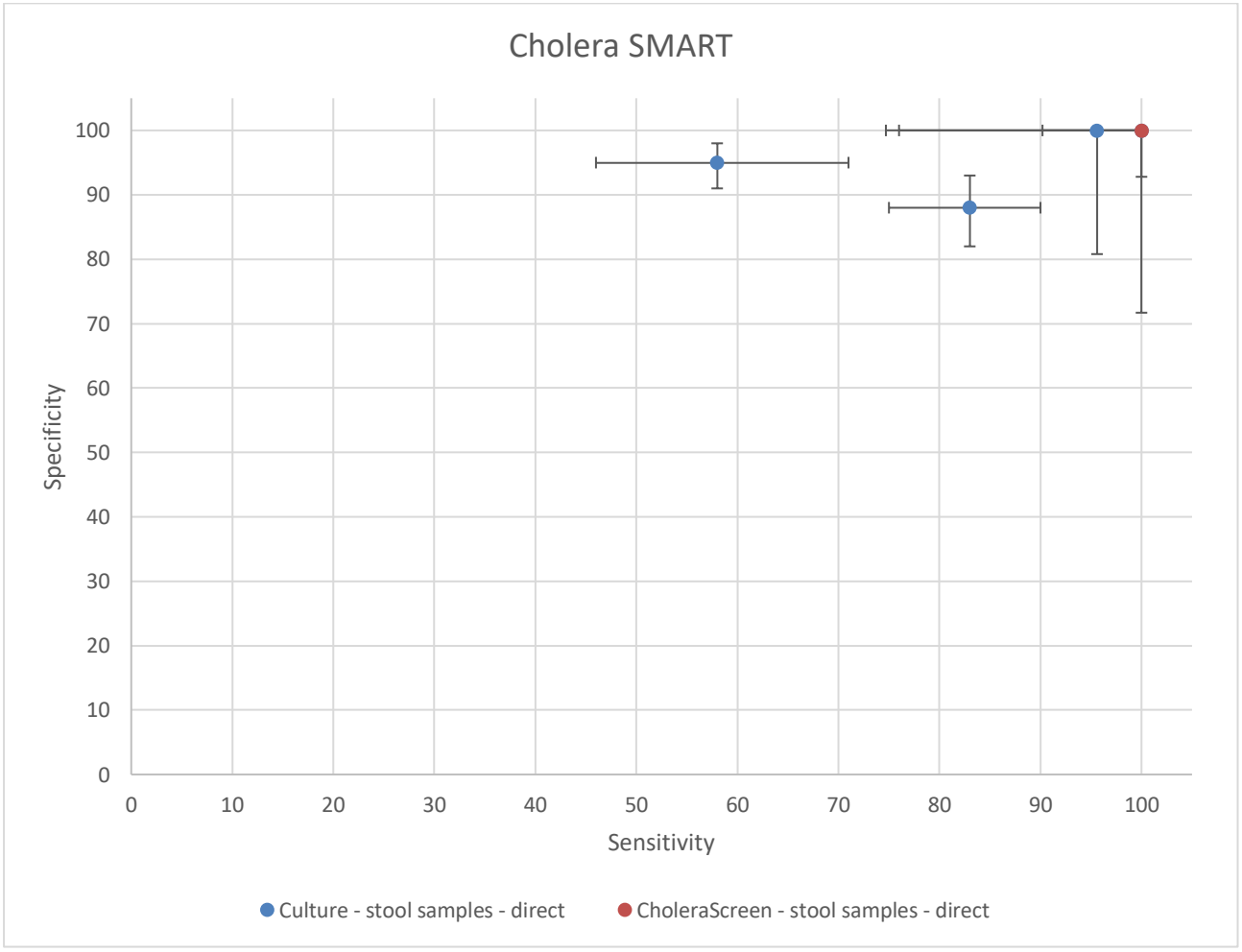
## Immunologically-based tests: CholeraSMART

*Table 2. Sensitivity and specificity results for individual studies reporting on Cholera SMART*

| Test: Cholera SMART (New Horizons Diagnostics) | | | | |
|---|---|---|---|---|
| **Intended Location: Field** | | | | |
| **Target: Antigen A of V. cholerae O1 lipopolysaccharide** | | | | |
| **Reference test: Bacterial Culture** | | | | |
| **Sample type: Stool** | | | | |
| **Study ID** | **n** | **Enrichment details (if relevant)** | **Sensitivity** | **Specificity** |
| *Direct Samples* | | | | |
| Bolaños 2004 | 28 | | 100 (74.7-100) [1] | 100 (71.7-100) [1] |
| Hasan 1994b (Bangladesh) | 44 | | 95.6 (76-99.8) [1] | 100 (80.8-100) [1] |
| Kalluri 2006 (lab technicians) | 254 | | 83 (75-90) | 88 (82-93) |
| Kalluri 2006 (field technicians) | 212 | | 58 (46-71) | 95 (91-98) |
| **Reference test: Cholera Screen** | | | | |
| **Sample type: Stool** | | | | |
| *Direct Samples* | | | | |
| Hasan 1994b (Mexico) | 108 | | 100 (90.2-100) [1] | 100 (92.8-100) [1] |

[1] Confidence Intervals not provided; calculated from raw numbers

*Figure 2. Plot of sensitivities and specificities for individual studies. Each dot represents a single study result, sub-grouped by colour according to 'reference test – sample type – enrichment status' as per the legend below. Error bars show 95% Confidence Intervals*



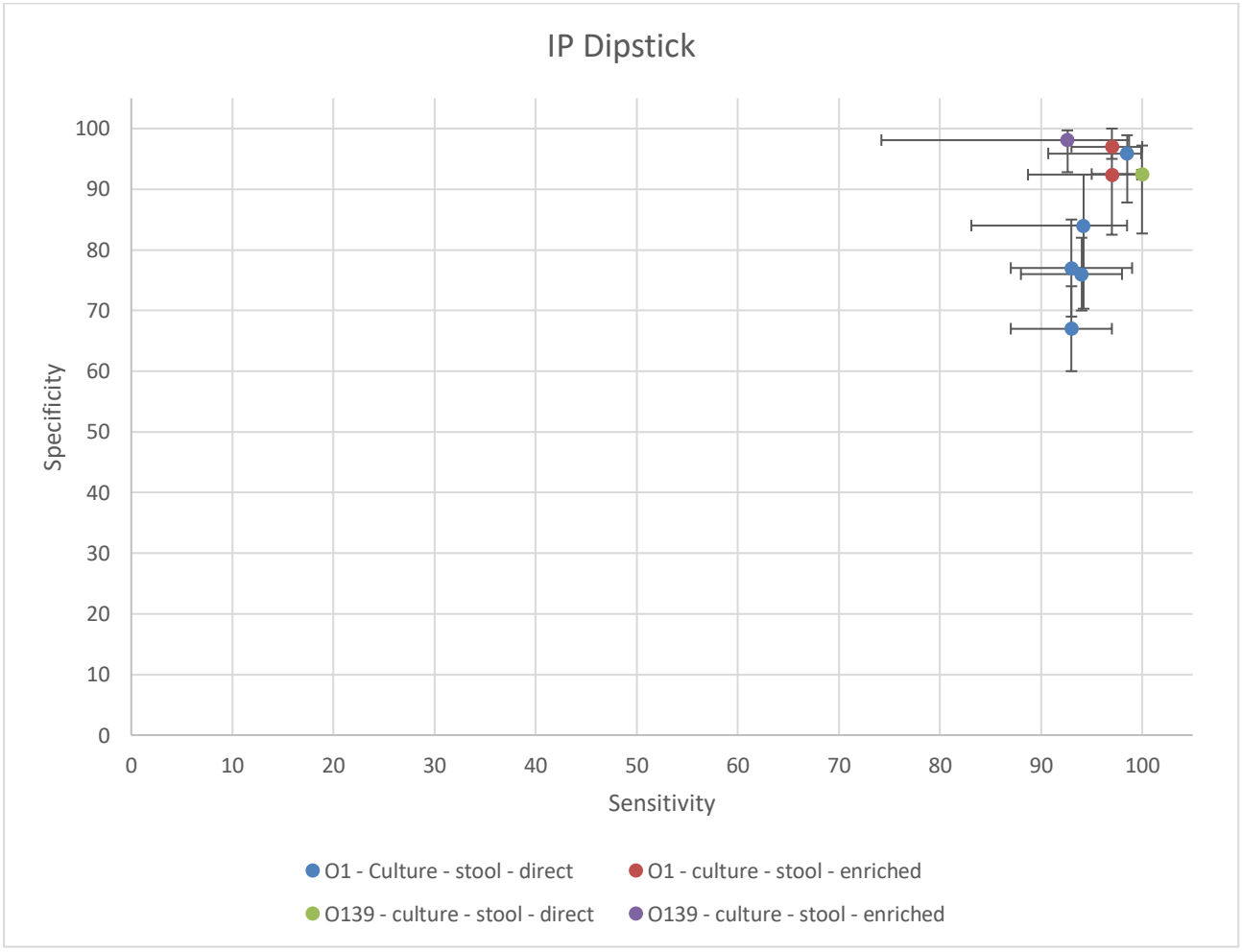
## Immunologically-based tests: IP Dipstick

*Table 3. Sensitivity and specificity results for individual studies reporting on IP Dipstick*

| Test: IP Dipstick (Institut Pasteur) | | | | |
|---|---|---|---|---|
| **Intended Location: Field** | | | | |
| **Target: V. cholerae O1 lipopolysaccharide** | | | | |
| **Reference test: Bacterial Culture** | | | | |
| **Sample type: Stool** | | | | |
| **Study ID** | **n** | **Enrichment details (if relevant)** | **Sensitivity** | **Specificity** |
| *Direct Samples* | | | | |
| Wang 2006 | 172 | | 93 (87-99) | 77 (69-85) |
| Kalluri 2006 (field technicians) | NR | | 93 (87-97) | 67 (60-74) |
| Kalluri 2006 (lab technicians) | NR | | 94 (88-98) | 76 (70-82) |
| Nato 2003 (Madagascar) | 140 | | 98.5 (90.7-99.9) [1] | 95.9 (87.8-98.9) [1] |
| Nato 2003 (Bangladesh) | 102 | | 94.2 (83.1-98.5) [1] | 84 (70.3-92.4) [1] |
| *Enriched Samples* | | | | |
| Wang 2006 | 219 | 6h in APW | 97 (93-100) | 97 (95-100) |
| Bhuiyan 2003 | 133 | 4h in APW | 97 (88.7-99.5) [1] | 92.4 (82.5-97.2) [1] |
| **Target: V. cholerae O139 lipopolysaccharides** | | | | |
| **Reference test: Bacterial Culture** | | | | |
| **Sample type: Stool** | | | | |
| *Direct Samples* | | | | |
| Nato 2003 (Bangladesh) | 158 | | 100 (95-100) [1] | 92.5 (82.7-97.2) [1] |
| *Enriched Samples* | | | | |
| Bhuiyan 2003 | 134 | 4h in APW | 92.6 (74.2-98.7) [1] | 98.1 (92.8-99.7) [1] |

[1] Confidence Intervals not provided; calculated from raw numbers

*Figure 3. Plot of sensitivities and specificities for individual studies. Each dot represents a single study result, sub-grouped by colour according to 'serotype - reference test – sample type – enrichment status' as per the legend below. Error bars show 95% Confidence Intervals*



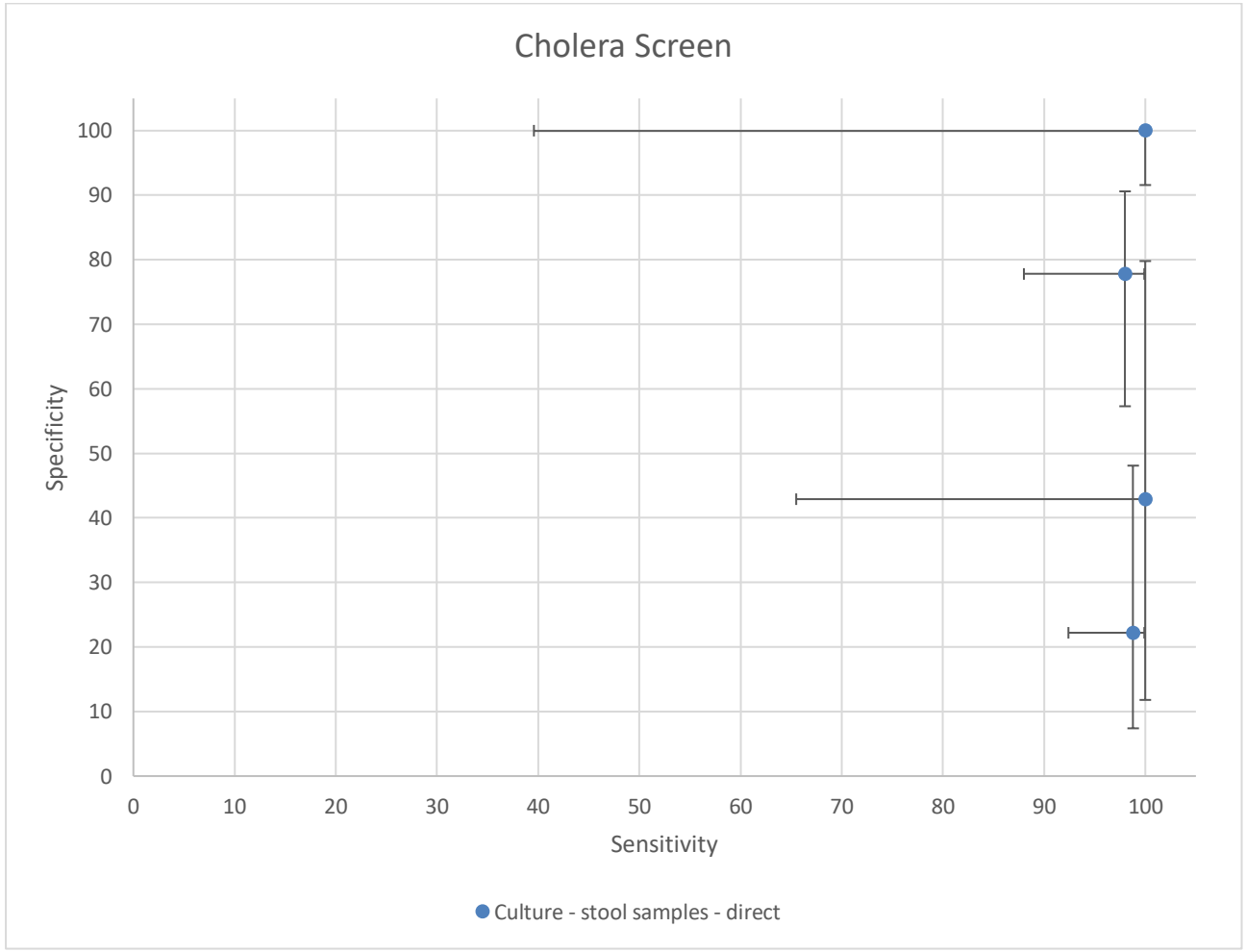
## Immunologically-based tests: Cholera Screen

*Table 4. Sensitivity and specificity results for individual studies reporting on Cholera Screen*

| **Test: Cholera Screen** | | | | |
|---|---|---|---|---|
| **Intended Location: Field** | | | | |
| **Target: 'A' factor of V. cholerae lipopolysaccharide O1** | | | | |
| **Reference test: Bacterial Culture** | | | | |
| **Sample type: Stool** | | | | |
| **Study ID** | **n** | **Enrichment details (if relevant)** | **Sensitivity** | **Specificity** |
| *Direct Samples* | | | | |
| Colwell 1992 (Guatemala) | 17 | | 100 (65.5-100) [1] | 42.9 (11.8-79.8) [1] |
| Colwell 1992 (Bangladesh) | 77 | | 98 (88-99.9) [1] | 77.8 (57.3-90.6) [1] |
| Carillo 1994 | 99 | | 98.8 (92.4-99.9) [1] | 22.2 (7.4-48.1) [1] |
| Islam 1994 | 57 | | 100 (39.58-100) [1] | 100 (91.58-100) [1] |

[1] Confidence Intervals not provided; calculated from raw numbers

*Figure 4. Plot of sensitivities and specificities for individual studies. Each dot represents a single study result, sub-grouped by colour according to 'reference test – sample type – enrichment status' as per the legend below. Error bars show 95% Confidence Intervals*



## Immunologically-based tests: ELISA-based tests

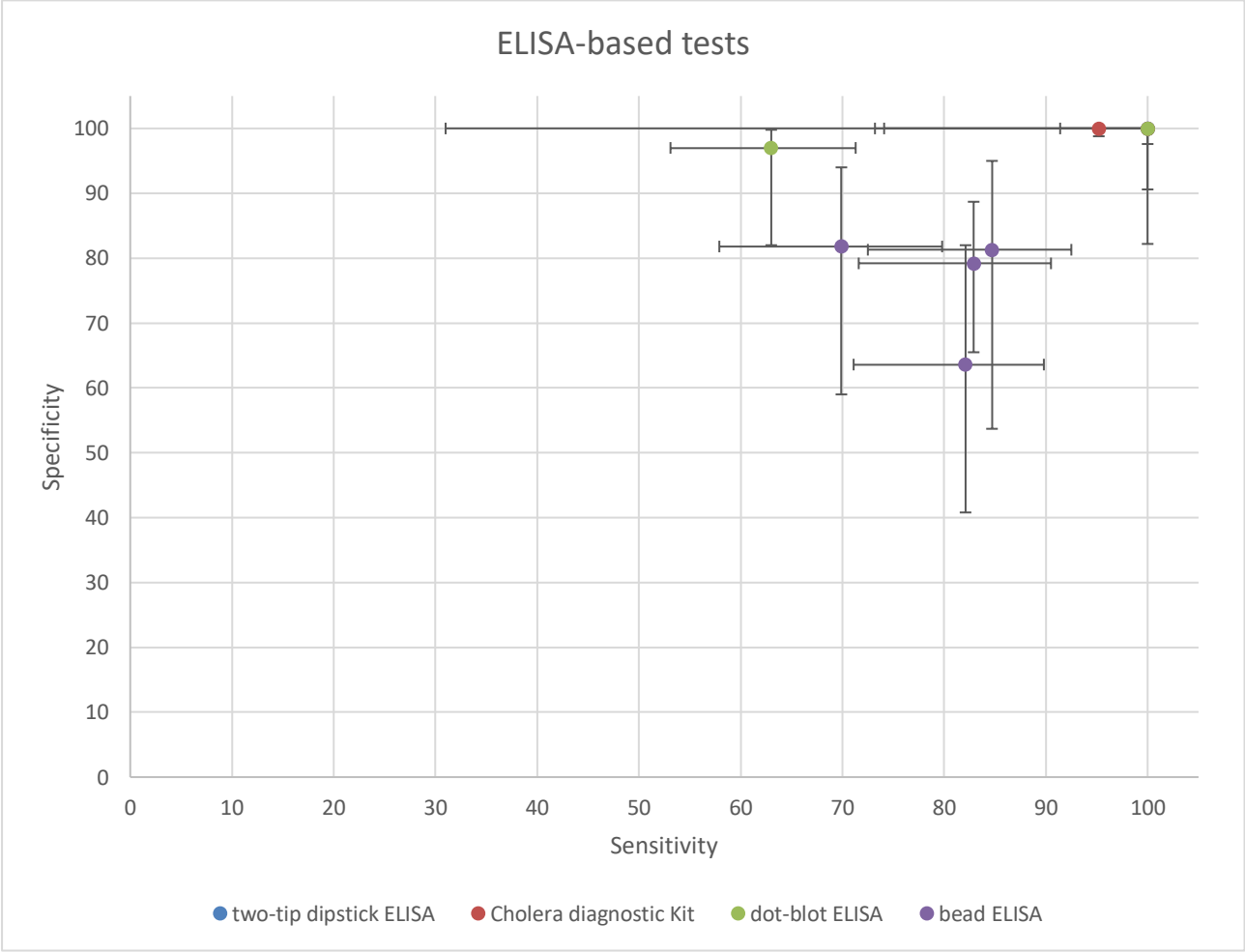
*Table 5. Sensitivity and specificity results for individual studies reporting on ELISA-based tests*

| Study ID | n | Enrichment and sub-group details (if relevant) | Sensitivity | Specificity |
|---|---|---|---|---|
| **Test: ELISA-based tests - two-tip dipstick sandwich ELISA** | | | | |
| **Intended Location: field** | | | | |
| **Target: V. cholerae O1 and O139 (Stool); Ctx b and ompw of O1 and O139 (water)** | | | | |
| **Reference test: Culture** | | | | |
| **Sample type: Stool or water** | | | | |
| *Enriched Samples* | | | | |
| Tuteja 2007 (Stool samples) | 75 | 4h in APW | 100 (91.4-100) [1] | 100 (82.2-100) [1] |
| Tuteja 2007 (Water samples) | 50 | 4h in APW | 100 (31-100) [1] | 100 (90.6-100) [1] |
| **Test: ELISA-based tests - Cholera diagnostic kit (Mab-based dot-blot ELISA)** | | | | |
| **Intended Location: field** | | | | |
| **Target: V cholerae O1 antigen A** | | | | |
| **Reference test: Culture** | | | | |
| **Sample type: Stool** | | | | |
| *Enriched Samples* | | | | |
| Supawat 1994 (Diarrhoeic patients) | 211 | 4h in APW | 100 (73.2-100) [1] | 100 (97.6-100) [1] |
| Supawat 1994 (Household contacts) | 415 | 4h in APW | 95.2 (74.1-99.8) [1] | 100 (98.8-100) [1] |
| **Test: ELISA-based tests - dot-blot ELISA** | | | | |
| **Intended Location: laboratory** | | | | |
| **Target: V. cholerae O1 antigen or V. cholerae O139 antigen** | | | | |
| **Reference test: Culture** | | | | |
| **Sample type: Stool** | | | | |
| *Direct Samples* | | | | |
| Chaicumpa 1995 | 147 | O1 antigen | 63 (53.1-71.3) [1] | 97 (82-99.8) [1] |

| | | | | |
|---|---|---|---|---|
| *Enriched Samples* | | | | |
| Chaicumpa 1998 | 6497 | O139 antigen | 100 (89.6-100) [1] | 99.95 (99.8-99.9) [1] |
| **Test: ELISA-based tests - bead ELISA** | | | | |
| **Intended Location: laboratory** | | | | |
| **Target: Cholera toxin** | | | | |
| **Reference test: Culture** | | | | |
| **Sample type: Stool** | | | | |
| *Direct Samples* | | | | |
| Ramamurthy 1996 (PAb-based) | 95 | | 82.1 (71.1-89.8) [1] | 63.6 (40.8-82) [1] |
| Ramamurthy 1996 (MAb-based) | 95 | | 69.9 (57.9-79.8) [1] | 81.8 (59-94) [1] |
| Ramamurthy 1993 | 123 | | 82.9 (71.6-90.5) [1] | 79.2 (65.5-88.7) [1] |
| Ramamurthy 1992 | 75 | | 84.7 (72.5-92.5) [1] | 81.3 (53.7-95) [1] |

[1] Confidence Intervals not provided; calculated from raw numbers

*Figure 5. Plot of sensitivities and specificities for individual studies. Each dot represents a single study result, sub-grouped by colour according to type of ELISA test, as per the legend below. Error bars show 95% Confidence Intervals*



## Other tests

*Table 6. Sensitivity and specificity results for individual studies reporting on other tests not mentioned above*

| Test: Other tests | | | | | | |
|---|---|---|---|---|---|---|
| **Intended Location: Field** | | | | | | |
| **Reference test: Culture** | | | | | | |
| **Sample type: Stool** | | | | | | |
| **Study** | **Test name** | **Target** | **n** | **Enrichment and sub-group details (if relevant)** | **Sensitivity** | **Specificity** |
| *Direct Samples* | | | | | | |
| Kalluri 2006 | Medicos Cholera Dip Stick | Unknown | 304 | Field technicians | 84 (77-91) | 79 (73-85) |
| Kalluri 2006 | Medicos Cholera Dip Stick | Unknown | 303 | Lab technicians | 88 (81-94) | 80 (73-95) |
| Bolaños 2004 | PDK (Pathogen Detection Kit) | Antigen A of V. cholerae O1 lipopolysaccharide | 27 | High probability specimens | 100 (71.7-100) [1] | 86 (56.2-97.5) [1] |
| Qadri 1995 | Bengal SMART | A' factor of V. cholerae O139 | 189 | | 97 (89.5-99.1) [1] | 100 (95.5-100) [1] |
| Hasan 1995 | BengalScreen | lipopolysaccharide antigen of V. cholerae O139 | 35 | | 95 (71.9-99.7) [1] | 100 (75.9-100) [1] |
| Hasan 1995 | Bengal DFA | lipopolysaccharide antigen of V. cholerae O139 | 35 | | 100 (79.1-100) [1] | 100 (75.9-100) [1] |
| Qadri 1994 | co-agglutination test (COAT) | lipopolysaccharide antigens of V. cholerae O139 | 230 | | 92 (84.1-96.5) [1] | 100 (96.7-100) [1] |
| Carillo 1994 | latex agglutination test | A' factor of V. cholerae lipopolysaccharide O1 | 100 | | 100 (94.4-100) [1] | 33 (14.4-58.8) [1] |
| Islam 2019 | Cholkit | lipopolysaccharide antigen of V. cholerae O1 | 1355 | | 79.4 (62.1-91.3) | 87.4 (85.5-89.1) |
| Sayeed 2018 | Cholkit | lipopolysaccharide antigen of V. cholerae O1 | 76 | | 97.7 (88.4-99.9) [2] | 96.5 (88.6-99.6) [2] |
| Matias 2017 | Artron RDT | Vibrio cholerae O1 and O139 | 129 | | 98.6 (92.7-100) | 69.1 (55.2-80.9) |
| Matias 2017 | SD Bioline | V. choleraeO1 and O139 antigens | 451 | | 81.1 (75.6-85.8) | 92.8 (88.4-95.9) |

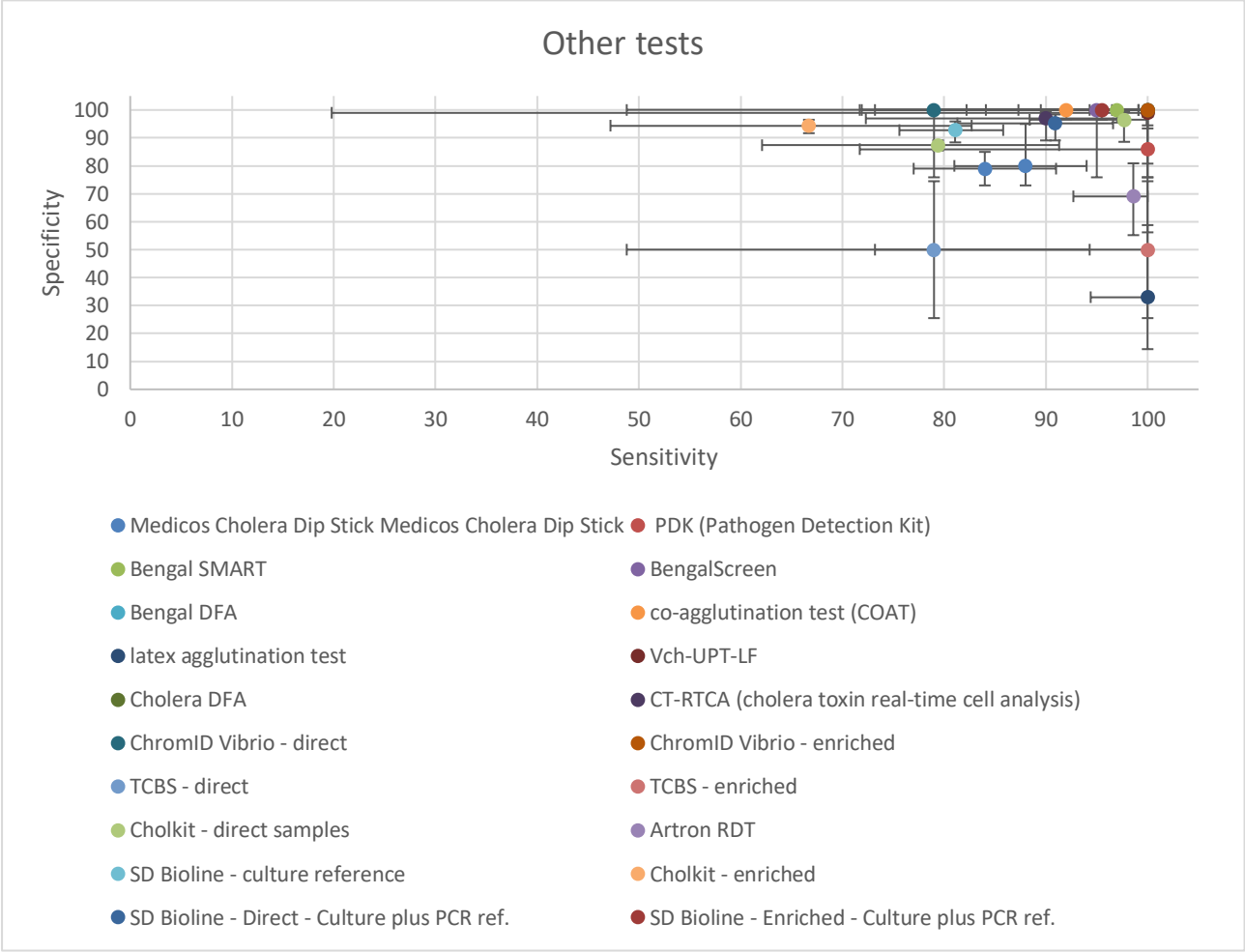
| | | | | | | |
|---|---|---|---|---|---|---|
| *Enriched Samples* | | | | | | |
| Islam 2019 | Cholkit | lipopolysaccharide antigen of V. cholerae O1 | 424 | | 66.7 (47.2-82.7) | 94.4 (91.7-96.5) |
| **Reference test: Culture plus PCR** | | | | | | |
| **Sample type: Stool** | | | | | | |
| *Direct Samples* | | | | | | |
| Mwaba 2018 | SD Bioline | V. choleraeO1 and O139 antigens | *170* | | 90.9 (81.3-96.6) | 95.2 (89.1-98.4) |
| *Enriched Samples* | | | | | | |
| Mwaba 2018 | SD Bioline | V. choleraeO1 and O139 antigens | *170* | | 95.5 (87.3-99.1) | 100 (96.5-100) |
| **Reference test: Combination** | | | | | | |
| **Sample type: Water** | | | | | | |
| *Direct Samples* | | | | | | |
| Hao 2017 | Vch-UPT-LF | V. cholerae O1 or O139 | 96 | O1 | 100 (71.7-100) [1] | 100 (94.5-100) [1] |
| Hao 2017 | Vch-UPT-LF | V. cholerae O1 or O139 | 96 | O139 | 100 (19.8-100) [1] | 99 (93.4-99.9) [1] |
| **Test: Other tests** | | | | | | |
| **Intended Location: Laboratory** | | | | | | |
| **Reference test: Culture** | | | | | | |
| **Sample type: Stool** | | | | | | |
| *Direct Samples* | | | | | | |
| Hasan 1994a | Cholera DFA | A' factor of V. cholerae lipopolysaccharide O1 | 44 | | 100 (82.2-100) [1] | 100 (80.8-100) [1] |
| **Reference test: Combination** | | | | | | |
| **Sample type: Stool** | | | | | | |
| *Direct Samples* | | | | | | |
| Jin 2013 | CT-RTCA (cholera toxin real-time cell analysis) | Cholera toxin | 100 | | 90 (72.3-97.4) [1] | 97 (89.1-99.5) [1] |
| Eddabra 2011 | ChromID Vibrio | V. cholerae bacterial strains | 30 | | 79 (48.8-94.3) [1] | 100 (75.9-100) [1] |

| | | | | | | |
|---|---|---|---|---|---|---|
| Eddabra 2011 | TCBS | V. cholerae bacterial strains | 30 | | 79 (48.8-94.3) [1] | 50 (25.5-74.5) [1] |
| *Enriched Samples* | | | | | | |
| Eddabra 2011 | ChromID Vibrio | V. cholerae bacterial strains | *30* | Yes - 5-8h in APW | 100 (73.2-100) [1] | 100 (75.9-100) [1] |
| Eddabra 2011 | TCBS | V. cholerae bacterial strains | 30 | Yes - 5-8h in APW | 100 (73.2-100) [1] | 50 (25.5-74.5) [1] |

[1] Confidence Intervals not provided; calculated from raw numbers

[2]Result estimated using Bayesian Latent Modelling

*Figure 6. Plot of sensitivities and specificities for individual studies. Each dot represents a single study result, sub-grouped by colour according to type test, as per the legend below. Error bars show 95% Confidence Intervals*



## PCR-based tests

*Table 7. Sensitivity and specificity results for individual studies reporting on PCR-based tests*

| Test: PCR-based tests | | | | | | |
|---|---|---|---|---|---|---|
| **Intended Location: laboratory** | | | | | | |
| **Reference test: Culture** | | | | | | |
| **Sample type: Stool** | | | | | | |
| **Study** | **Test name** | **Target** | **n** | **Enrichment and sub-group details (if relevant)** | **Sensitivity** | **Specificity** |
| *Direct Samples* | | | | | | |
| Albert 1997 | PCR assay (with new primers O139-1 and O139-2) | V. cholerae O139 | 180 | Undiluted sample | 94 (84.7-98.1) [1] | 100 (95.9-100) [1] |
| Sayeed 2018 | Multiplex PCR | V. cholerae O1 and O139 specific rfb genes and cholera toxin gene ctxA | 76 | | 73.6 (58.5-85.7) [2] | 97.2 (93.2-99.2) [2] |
| *Enriched Samples* | | | | | | |
| Hoshino 1998 | Multiplex PCR (with: O139-rfb primers - O139F-2, O139R-2; O1-rfb primers - O1F-2, O1R-2; cholera toxin primers - VCT1, VCT2) | V. cholerae O1 and O139 rbf-specific genes and the ctxA gene. | 121 | 18h in APW | 100 (88.6-100) [1] | 95 (87.5-98.4) [1] |
| Ramamurthy 1993 | PCR (with primers specific to V. cholera O1) | Cholera Toxin gene of V. cholerae O1. | 123 | 18h in APW | 100 (93.5-100) [1] | 55 (40.6-68.2) [1] |
| **Reference test: Culture** | | | | | | |
| **Sample type: Water** | | | | | | |
| *Enriched Samples* | | | | | | |
| Momtaz 2013 | PCR (with primers specific to espM gene) | epsM gene of V. cholerae | 448 | 6-8h in APW | 100 (31-100) [1] | 100 (98.9-100) [1] |
| **Reference test: Conventional assay** | | | | | | |
| **Sample type: Stool** | | | | | | |
| *Direct Samples* | | | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| Liu 2013 | Taqman array Card | toxR gene | 80 | | 100 (59.8-100) [1] | 100 (93.7-100) [1] |
| **Reference test: PCR Luminex** | | | | | | |
| **Sample type: Stool** | | | | | | |
| *Direct Samples* | | | | | | |
| Liu 2013 | Taqman array Card | toxR gene | 109 | | 100 (62.9-100) [1] | 100 (95.4-100) [1] |

[1] Confidence Intervals not provided; calculated from raw numbers

[2] Result estimated using Bayesian Latent Modelling

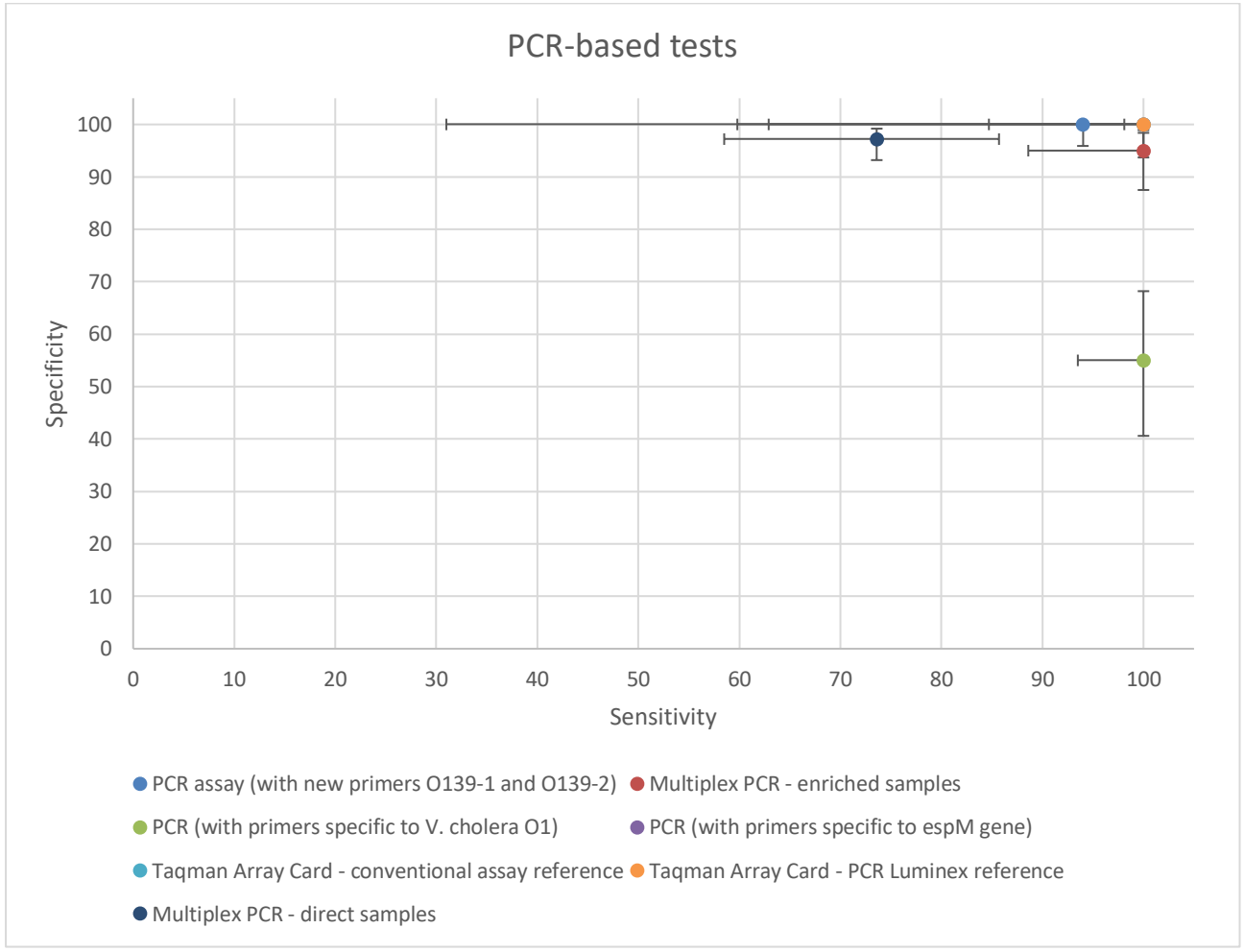



*Figure 7. Plot of sensitivities and specificities for individual studies. Each dot represents a single study result, sub-grouped by colour according to type PCR test and reference or enrichment status, as per the legend below. Error bars show 95% Confidence Intervals.*